PROBLEM STATEMENT

Automated reading of TLC plates

What the system receives, what it can extract, and what that enables

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What is TLC (one line)

Thin Layer Chromatography: a drop of sample on a coated plate, a solvent carries its components up at different speeds, and they separate into spots — one plate visually shows what compounds are in the sample.

THE SYSTEM AT A GLANCE

INPUT

Photos of 4-lane TLC plates

- S · Co · R · Sd lanes
- ID, time and solvent
- written on the plate
- imaged under UV

→

READ

Every signal on the plate

- spot position (Rf)
- spot count
- spot intensity
- spot shape

→

EXTRACT

Measured values

- which spot is which
- SM left in R lane
- product formed
- extra spots flagged

→

DECIDE

Recorded outcomes

- complete / incomplete
- conversion estimate
- by-product warning
- same call every time

THE GAP TODAY

Every plate is read by eye under a UV lamp. The reading is subjective — two chemists can quote 50% vs 70% conversion from the same plate. It is qualitative (“the spot looks lighter”, never a number) and it is not recorded in any structured way — the information on a plate is lost once the plate is discarded. The decisions that determine purity and yield rest on whoever is holding the plate.

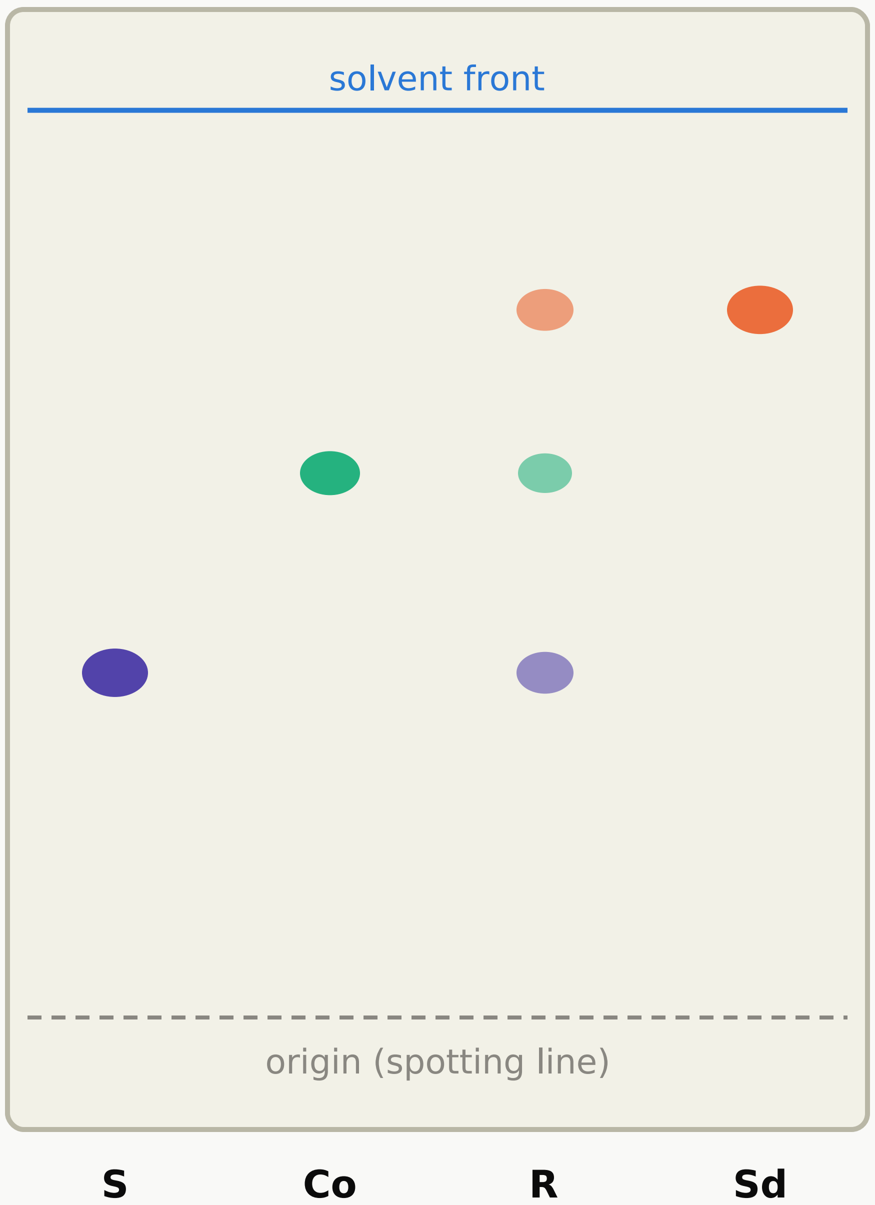
Scope: automate the read-out of the plate. The chemistry stays with the chemist.

1 · INPUT

What the system receives

The plate format the lab already uses — four lanes, metadata written on the plate

MEHQ-P31
4 hr (30% EA-Hex)



S — starting material
Pure reference. Fixes where the SM spot sits, so its fading can be tracked in the R lane.

Co — reactants
The other reactants' reference positions, so their spots are never mistaken for something new.

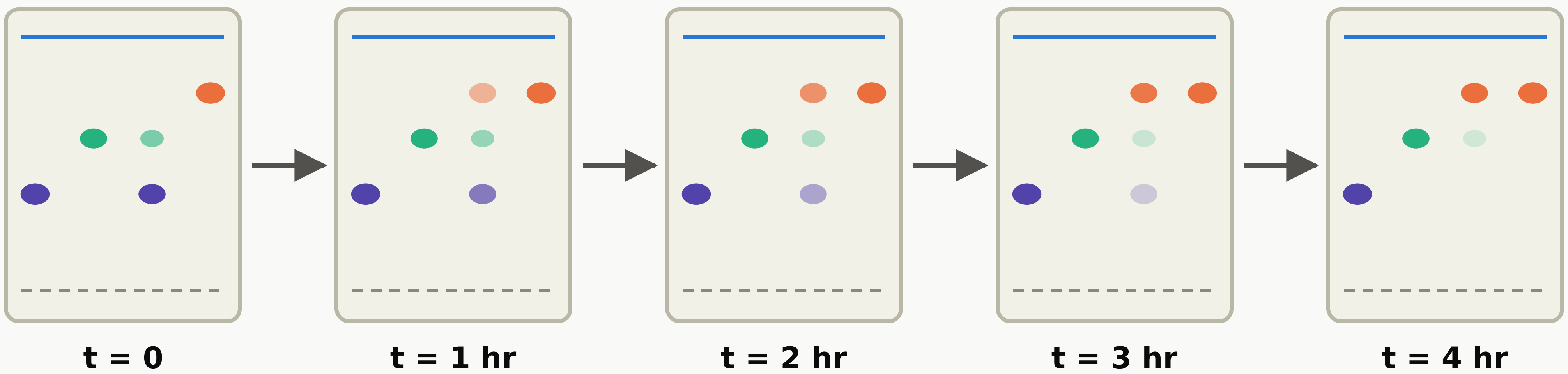
R — reaction mixture
A drop pulled from the flask. The lane that actually changes — the one that gets read.

Sd — standard product
Authentic product sample. An R-lane spot at this height means the product is forming.

METADATA — ALREADY HANDWRITTEN ON EVERY PLATE

reaction ID (MEHQ-P31) | time sampled (4 hr) | mobile phase (30% EA-Hex) | imaging (UV 254 nm)

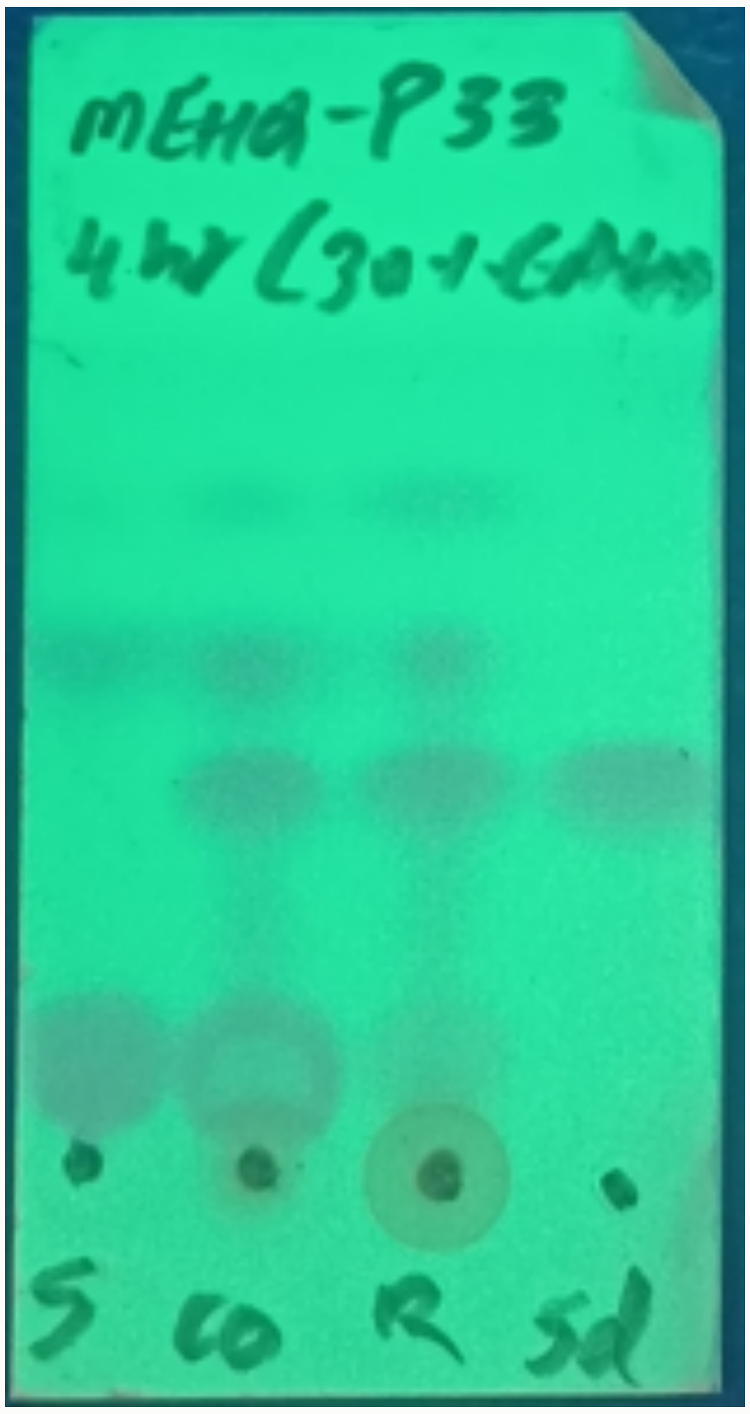
TODAY: ONE PLATE NEAR THE END. THE SAME FORMAT AT INTERVALS UNLOCKS MORE (PAGE 5)



In the R lane the SM spot (violet) fades and the product spot (orange) grows. Each plate is one frame; a series of frames is what turns snapshots into a picture of how the reaction moved.

The actual input, as photographed in our lab

UV 254 nm images from analytics — exactly what the system would receive



Handwritten header — reaction ID, time sampled, mobile phase (e.g. MEHQ-P33, 4 hr, 30% EA-Hex). Metadata rides on the plate itself.

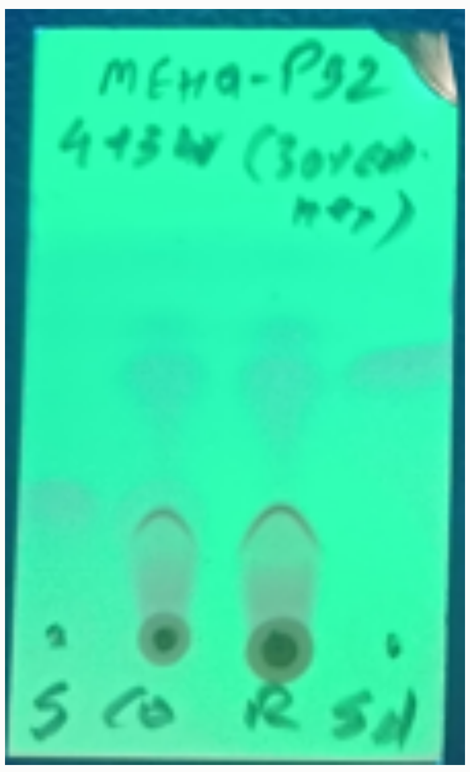
Developed spots — dark on the green UV-254 background. Height gives Rf; darkness gives relative amount; one row per compound.

Baseline residues — material that never moved: very polar components or sample left at the origin.

Lane labels S · Co · R · Sd — written under the lanes at the bottom edge.

MORE PLATES FROM THE ARCHIVE

P31 · 4 hr



P32 · 4+3 hr

P30 · 4 hr

P32-1 · 3 hr (ACN)

Note: P32 was checked at 4 hr and again after 3 more hours — the same plate format at two timepoints. That is exactly the series the time analysis on page 5 builds on.

What each signal on the plate tells us

Five independent signals, read from the same image

1 POSITION — Rf

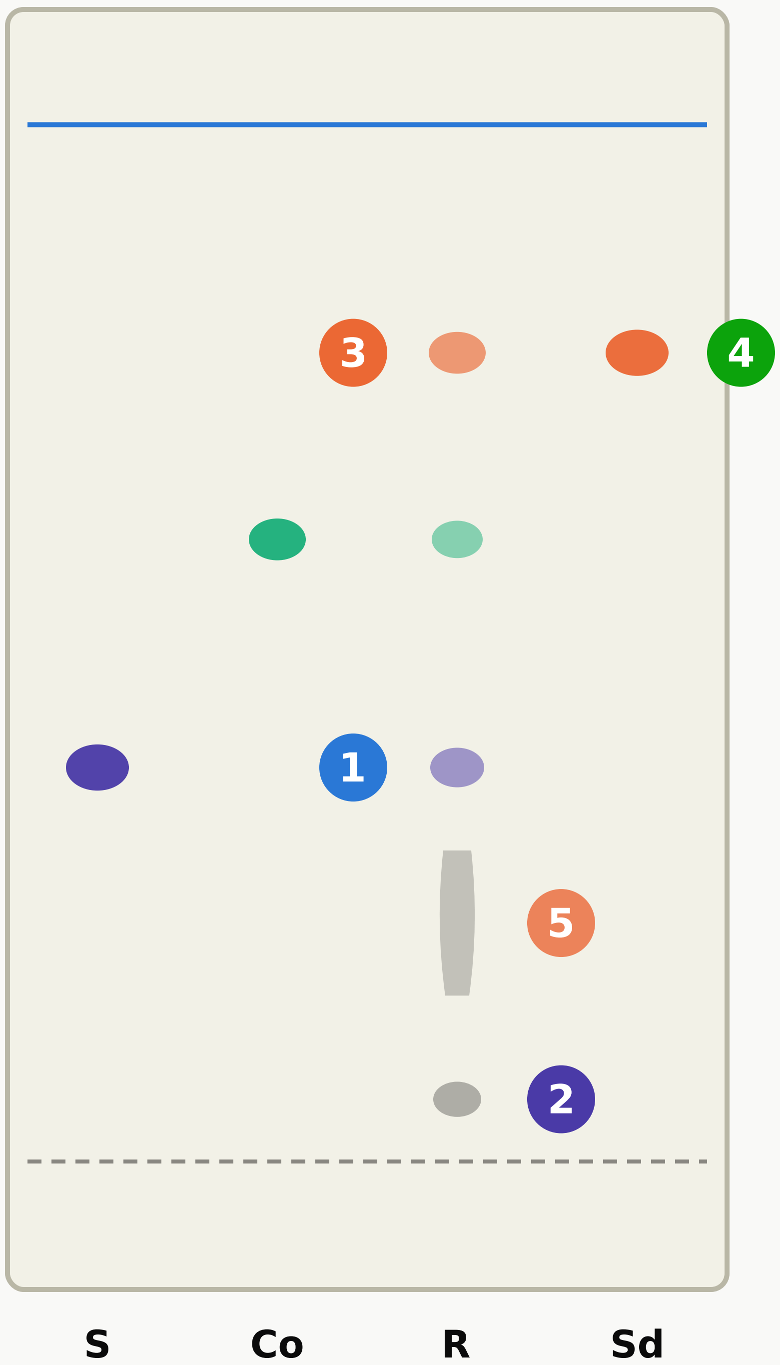
How far a spot travelled ÷ how far the solvent went. Matches a spot to the S / Co / Sd references, and encodes its polarity.

2 COUNT — purity

How many spots the R lane holds. Extra spots = by-products forming. One clean spot at the end = a clean, scalable route.

5 SHAPE & BASELINE

Streaks = overloaded sample or decomposition. Spots stuck at the origin = very polar material accumulating.



3 INTENSITY — amount

Spot darkness × size tracks how much compound is present. Read against reference spots it becomes % remaining / % formed.

4 MATCH vs Sd LANE

R-lane spot at exactly the standard product's height = the product is confirmed forming — not just “some new compound”.

SIGNAL → INFORMATION → DECISION

Signal on the plate	What it tells us	What it decides
Spot position (Rf)	which compound each spot is	identify SM / product / newcomers
Spot count	how clean the reaction is	flag by-products early, judge the route
Spot intensity	how much — % conversion	continue, intervene, or stop
Match vs Sd lane	product confirmed	trust the conversion estimate
Spot shape	sample & technique health	trust or re-run the plate

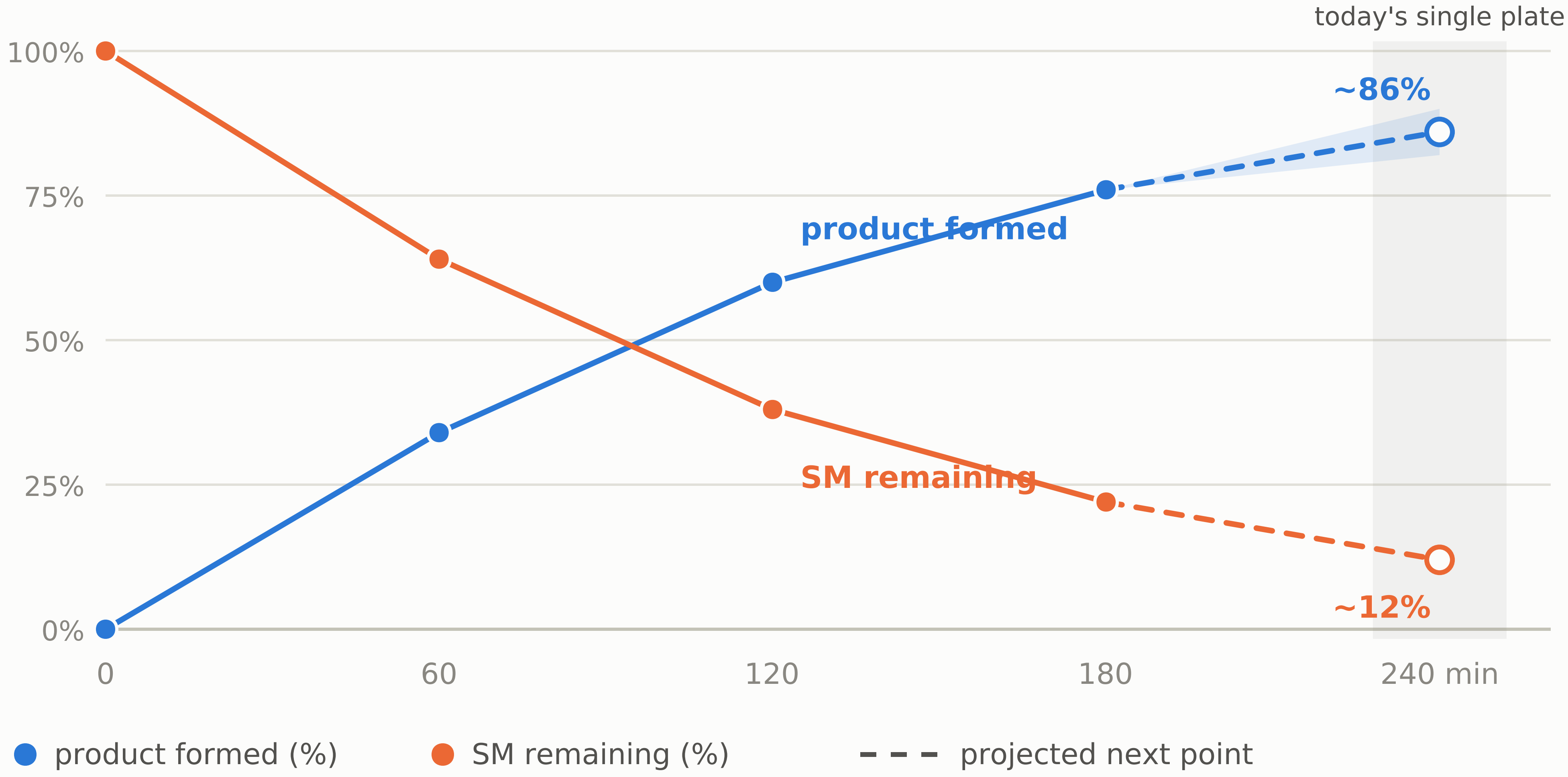
4 · TIME

What interval plates would add

Today one plate is run near the end (~4 hr). Plates at intervals carry more information.

Conversion vs time, measured from spot intensities (illustrative)

dots = interval plates · dashed = projection to the next timepoint · today only the last point exists



SINGLE END-POINT PLATE vs INTERVAL PLATES

End-point plate

Done or not done at 4 hr. Whether the product formed. Whether extra spots exist. One point on the curve — no rate, no history.

Intervals: + rate

How fast SM is being consumed — fast, slow, or flattening. Whether the reaction is approaching completion or stalling.

Intervals: + timing

When conversion stops improving — the earliest safe stop point, instead of a fixed 4 hr for every reaction.

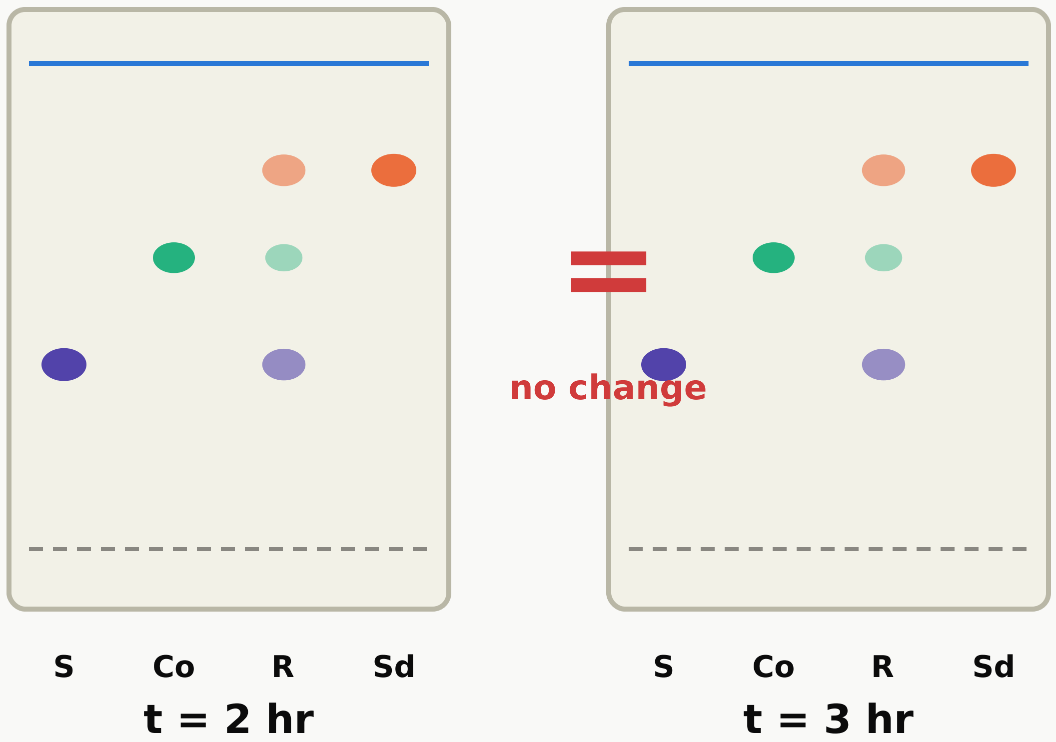
Intervals: + history

When a by-product first appeared and how it grew. Stored runs become comparable across reactions of the same route.

The system works with either input. With today's end-point plates it standardises the completion and purity read-out; if interval plates are taken, the same reading produces the curve above.

How we know a reaction is not moving forward

The signal is the absence of change — consecutive plates that look the same

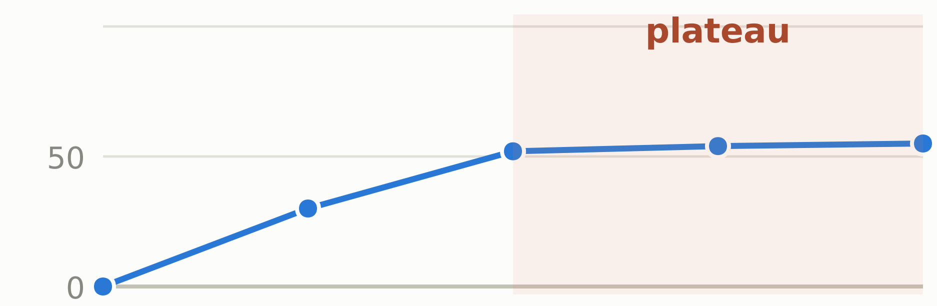


Nothing changed in a full hour

SM spot in R lane: same intensity as before. Product spot: stopped growing. No new spots appearing either.

By eye this is a judgment call. Measured as numbers, it is a flat line: ΔSM ≈ 0 and Δproduct ≈ 0 across consecutive plates — unambiguous, and recorded.

The same stall, as a curve



product % flattens at ~55% — conversion has stopped short

THE DECISION EVERY NEW PLATE FEEDS

New plate at time t: measure SM % and product % in R lane → **Compare with the previous plates: ΔSM, Δproduct, any new spots**

PROGRESSING

SM still falling, product still rising → let it run; TLC again at the next interval.

STALLED

No change across ~2 intervals, SM still present → intervene: temperature / reagent charge / catalyst. Keep monitoring.

COMPLETE

SM spot gone (or below limit) → stop: quench and work up. Running longer only breeds by-products.

What a plate cannot say

Knowing the limits is part of understanding the tool

Same height ≠ same compound

Two different compounds can land at the same Rf. The Sd-lane match is strong evidence, not absolute proof — final identity comes from MS/NMR.

Some compounds are invisible

UV 254 only reveals conjugated molecules. UV-silent compounds need chemical stains — so the imaging condition (UV / stain) is part of every plate's data.

Rf drifts between days

Solvent ratio, humidity and plate batch shift absolute positions. Only same-plate comparisons are rigorous — which is exactly why all reference lanes ride on each plate.

Semi-quantitative by nature

Plate numbers are for in-process decisions, not release certificates. HPLC still owns the final purity/yield figures — this replaces the visual read-out, not the analytics lab.

SUMMARY

A TLC plate already answers the questions that decide purity and yield: what is in the flask, whether the product formed, roughly how much converted, whether extra spots appeared. Today those answers are read qualitatively, by eye, and lost when the plate is discarded. The system takes the same four-lane plates the lab already runs — S, Co, R, Sd, with the ID, time and solvent written on the plate — and returns the same answers as consistent, recorded measurements.

Method grounding: image-based TLC quantification is published and validated (TLCyzer, Nature Sci. Reports 2022; MIA-TLC reaction monitoring); the 4-lane layout is standard practice.